POC UPDATE

FIELDWORK

From natural language to reviewable environments

Project update for Dr. Melissa Cameron - October 28, 2026

Traceable plans. Reviewable assets. Measured next steps.

The goal and the foundation

FOUNDATION

- Goal: editable environments and adaptive emergency training.
- Backend owns accepted state, validation, logs and review.
- Two deterministic toy scenarios keep a repeatable baseline.
- Generated rooms and physics demos are separate evidence.

Not clinical guidance, protocol authority or trainee scoring.

The room studio works with a reviewed chair

ROOM STUDIO

- Hosted ARC planner creates bounded room plans.
- Seeded layouts, previews and persistent history.
- Blender builds editable rooms and renders.
- Unreal 5.8 bridge assembles maps and runs Chaos batches.

The user confirmed these workflows. Mixed furniture is the next integration.

Describe a design, prepare an asset

ASSET WORKFLOW

Description -> generated reference -> ARC Hunyuan shape/paint

-> retrieved model -> Blender MCP cleanup and convex proxies

-> editable .blend + Unreal FBX/material/collision package

- Frontend tracks jobs and reconnects to the same dispatch.
- Original source and provenance are retained.
- New designs remain review candidates.

Evidence has distinct levels

- Real generated chair, accepted repair and 19 convex pieces.
- User-confirmed chair, room and Chaos demonstrations.
- Native MCP/Blender/CoACD fixture: 28 closed hulls.
- 138 backend/ARC tests and frontend/CI checks passed.

The fixture is not the couch. Tests are not a live GPU or engine run.

The next live checks

NEXT CHECKS

- One-time SSH trust/agent setup and ARC checkout update.
- First automated couch and actual generated-model review.
- Generic Unreal import, materials and collision cooking.
- First live movement recording.

Then coordinate mixed-asset IDs, footprints and both assemblers.

Boundaries remain explicit

- Local single-user app; no production hosting or app login.
- Cleanup cannot reconstruct arbitrary missing/fused parts.
- Collision proxies and demo values are not calibrated physics.
- No sensing, projection or validated real-time adaptation.
- A generated room is not an approved training scenario.

Review the whole result

REVIEW + FEEDBACK

Show saved room layouts, the accepted chair and recorded evidence.

Show the new asset controls without claiming a running couch job.

Keep the deterministic session review as the repeatable fallback.

Feedback: Is the evidence clear? Which complete environment should come next?

Current setup and evidence: docs/README.md and docs/current-status.md